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Opsin gene and photopigment polymorphism in a prosimian primate

Gerald H. Jacobs [a,*], Jess F. Deegan II [b], Ying Tan [c,d], Wen-Hsiung Li [c]

[a] *Neuroscience Research Institute and Department of Psychology, University of California, Santa Barbara, CA 93106, USA*
[b] *Department of Psychology, California State University, Bakersfield, USA*
[c] *Department of Ecology and Evolution, University of Chicago, USA*
[d] *Department of Biology, University of Massachusetts, Boston, USA*

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Abstract

A recent genetic investigation found some species of prosimian to have an opsin gene polymorphism [Nature 402 (1999) 36]. In the present study the functional implications of this finding were explored in a correlated investigation of opsin genes and spectral sensitivity measurements of a diurnal prosimian, Coquerel's sifaka (*Propithecus verreauxi coquereli*). Spectra recorded using electroretinogram (ERG) flicker photometry reveal a cone photopigment polymorphism paralleling an opsin gene polymorphism detected by molecular methods. This species has two middle-to-long-wavelength cone pigments with peak sensitivities of about 545 and 558 nm and a short-wavelength-sensitive cone with a peak at about 430 nm. The distribution of these pigments among animals predicts the presence of both dichromatic and trichromatic forms of color vision. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Photopigments; Opsin genes; Color vision; Prosimians; Sifaka; *Propithecus verrauxi coquereli*

1. Introduction

Color vision in anthropoid primates reflects the presence of multiple classes of cone photopigment and a visual system organization that allows for neural comparisons of the signals generated by photon absorption in these different cone types. In recent years studies of the genes that specify photopigment opsins as well as measurements of cone photopigments and direct examinations of color vision have provided a good sketch of the nature of color vision among anthropoids. For instance, humans typically have three classes of cone photopigment with respective absorption peaks in the short (S), middle (M) and long (L) wavelengths. The genes specifying these three pigment opsins are located on chromosome 7 (S cone opsin) and at two neighboring locations on the X-chromosome (M and L cone opsins; Nathans, Thomas, & Hogness, 1986). The M and L opsin genes arose from a gene duplication assumed to have occurred following the split between platyrrhine and catarrhine primates (an event variously estimated to have occurred at times ranging from 35 to 60 million years ago; see Arnason, Gullberg, & Janke (1998)), but prior to catarrhine divergences. There are a number of well-studied polymorphic variations, but this general arrangement allows most individuals to have keen trichromatic color vision. Insofar as is now known, all other catarrhine primates have a very similar gene/photopigment arrangement and also enjoy trichromatic color vision (Jacobs & Deegan II, 1999).

The picture is quite different for the New World monkeys (Jacobs, 1998). These animals share with Old World primates an S-cone specified by an autosomal gene, but they have only a single X-chromosome opsin gene locus. Within various species there is polymorphic variation in the M/L opsin gene; frequently there are three alleles and, correspondingly, three versions of an M/L cone pigment. The retinas of male monkeys thus have only a single type of M/L pigment. In conjunction with the S cone this permits dichromatic color vision. Through X-chromosome inactivation heterozygous females produce two types of M/L cone and have trichromatic color vision. Among the platyrrhines, two exceptions to this pattern have been noted. In one case, that of the howler monkey (*Alouatta*), there has apparently been an X-chromosome opsin gene duplication yielding a photopigment/color vision arrangement much like described for catarrhine primates (Boissinot et al.,

* Corresponding author. Tel.: +1-805-893-2446; fax: +1-805-893-2005.
E-mail address: jacobs@psych.ucsb.edu (G.H. Jacobs).

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1997; Jacobs, Neitz, Deegan II, & Neitz, 1996; Kainz, Neitz, & Neitz, 1998). The other exception is the nocturnal owl monkey (*Aotus*). Owl monkeys have no polymorphic variation in their M/L opsin genes and, through mutational changes, have also lost functional S-cone pigment (Jacobs, Deegan II, Neitz, Crognale, & Neitz, 1993; Jacobs, Neitz, & Neitz, 1996). The result is that these monkeys have only a single cone pigment and lack a color vision capacity.

About a fifth of all of the primate species are prosimians. However, unlike the case of the anthropoids, there is only sparse information about color vision in prosimians. Two different biological arrangements have been detected. A nocturnal prosimian, the bushbaby (*Otolemur crassicaudatus*), was found to be similar the *Aotus* monkey, i.e., having a single M/L pigment in conjunction with a nonfunctional S-pigment gene (Jacobs et al., 1996). Two species of lemur (*Eulemur fulvus* and *Lemur catta*) were identified as having an S-pigment and a single M/L pigment. No polymorphism was detected in these animals suggesting they should all have the same type of dichromatic color vision (Jacobs & Deegan II, 1993). On the basis of this fragmentary evidence it seemed possible that trichromacy in primates appears only among the anthropoids, either as a result of an early X-chromosome gene duplication or through the presence of polymorphic variation in the M/L pigment genes. Recently, however, a genetic analysis conducted for 20 species of prosimian revealed the presence of allelic versions of M/L opsin genes in two diurnal species (Coquerel's sifaka and the red ruffed lemur; Tan & Li, 1999). The presence of such polymorphic variation predicts that heterozygous females of these species have the potential to achieve trichromatic color vision in the manner of female New World monkeys. To pursue this possibility further we have analyzed opsin genes and made measurements of the cone photopigments in the diurnal prosimian, Coquerel's sifaka.

2. Methods

2.1. Subjects

Adult sifakas (*Propithecus verreauxi coquereli*) of both sexes were studied. All were members of a colony maintained at the Duke University Primate Center, Durham, NC. For the genetic analysis, blood samples obtained from 19 animals were examined at the University of Chicago. From this group, a subset of 10 animals (five male, five female) served as subjects in a recording study conducted at Duke University. All the experimental procedures were in accord with the National Institutes of Health guidelines on the care and use of animals and were conducted according to protocols approved by the Duke University Institutional Animal Care and Use Committee. None of the subjects had any obvious ocular pathologies.

2.2. Genetic analysis

2.2.1. Sequencing exon five of the X-linked opsin gene

About 2 ml of blood was obtained by venipuncture from each of the animals. Genomic DNA was isolated from the blood sample using a Puregene kit (Gentra). Exon 5 of the X-linked opsin gene in each individual was amplified by the polymerase chain reaction (PCR) from the genomic DNA. The 5′primer sequence was 5′gtggcaaagcagcagaaagag3′ and the 3′primer sequence was 5′ctgccggttcataaagacgtagataat3′. The PCR conditions were: (1) one cycle for 3 min at 94°C for denaturation; (2) 30 cycles for 30 s at 94°C, 30 s at 54°C, 1 min at 72°C; (3) one cycle for 6 min at 72°C for final extension. PCR products were purified with the Wizard PCR purification system (Promega) and directly sequenced on an ABI 377 automatic sequencer.

2.2.2. Estimating λ_{max} of prosimian photopigments from opsin gene sequences

Previous studies of primate M and L cone opsin genes have identified three critical residue sites in exons 3, 4, and 5 that are responsible for the major portion of the spectral tuning of M/L photopigments: amino acid positions 180, 277 and 285, causing a spectral shift of ~5, 8 and 15 nm, respectively (Neitz, Neitz, & Jacobs, 1991; Shyue et al., 1998). The previous study showed that the first two critical sites are the same for the M and L opsin genes of prosimians (Ala, Tyr), and that the only critical residue differing between the two opsin genes is in exon 5 at position 285, with Ala for the M opsin and Thr for the L opsin (Tan & Li, 1999). Accordingly, only exon 5 was screened in this study. From the deduced amino acid sequence of exon 5, the λ_{max} of the photopigment can thus be estimated. If the opsin has Ala at critical site 285, the λ_{max} can be estimated to be the same as that of the marmoset P543 allele, i.e., ~543 nm (an M opsin); if the opsin has Thr at the site, its λ_{max} can be estimated to be $543 + 15 = 558$ nm (an L opsin) (Tan & Li, 1999). The sequences for exons 3–5 of the sifaka M/L opsins have been deposited in GenBank (GenBank accession AF431738–AF431743).

2.3. Recording apparatus and procedures

Inferences about the cone photopigments were derived from sensitivity measurements made using ERG flicker photometry. The recording system and general procedures employed for making these noninvasive electrophysiological measurements have both been described in detail so only a brief description is provided here (Jacobs & Deegan II, 1999; Jacobs, Neitz, & Krogh, 1996). ERGs were recorded using a Burian-style,

contact-lens electrode. In the flicker photometric procedure, ERGs elicited by a test light consisting of a train of square-wave pulses are electronically compared to the responses generated by an interleaved train of light pulses from a reference light. Signal filtering converts the ERG signals into sine waves tuned to the temporal frequencies of the stimuli, which are each (test and reference lights) modulated with a 25% duty cycle. To make a photometric equation the radiance of the test light is repetitively adjusted by changing the position of a neutral-density wedge, to the point where the ERG produced is equivalent to the signal produced by the reference light. Stimuli were derived from a three-beam optical system the output of which was imaged onto the retina in Maxwellian view (as a 57° circle). The test light came from a high-intensity monochromator (10 nm half-bandwidth). The reference light, and a third beam that served to provide adaptation lights, originated from tungsten-halide lamps. Neutral-density step filters and interference filters were used to control the content of the latter two beams.

Animals were initially anesthetized with 5% isoflurane delivered through a mask, then intubated and maintained on 1.5–2.0% isoflurane for the duration of the recording session. Depth of anesthesia was evaluated by continuous monitoring of heart rate, respiratory rate, body temperature, and expired CO_2. The cornea of one eye was anesthetized by topical application of proparacaine hydrochloride (0.5%) and the pupil dilated with mydriacyl (0.5% tropicimide). The ERG electrode was installed and the animal was positioned for recording using a specially designed head holder. Recordings were conducted in a surgery room with ambient lighting kept at a photopic level.

Flicker photometric equations were obtained by averaging the ERG responses to the last 50 cycles of a total of 70 stimulus cycles. During the course of the experiment equations for each of the test light/reference light combinations were made twice and the two values subsequently averaged. In ERG flicker photometry most repeat equations agree to a precision of 0.03 log unit or less (Jacobs et al., 1996). In the small number of cases where the two equations differed by more than 0.04 log unit a third equation was obtained and this was included in the final computation. Three separate sets of measurements were made.

(1) Complete spectral sensitivity functions were obtained from all the animals. The stimulus rate was 31.25 Hz. Photometric equations were made between monochromatic test lights and an achromatic (2450 K) reference light having a retinal illuminance of 3.3 log td. The test light was varied in 10 nm steps over the range from 440 or 450 to 650 or 660 nm, the end points depending on the sensitivity of the individual subject.

(2) A test was run on each subject to assess whether the retina contained one or more types of cone photopigment with maximum sensitivity in the middle- to long-wavelengths. This test serves to predict whether the animal should have dichromatic or trichromatic color vision (Jacobs & Neitz, 1987). ERG flicker photometric equations were made between a 540 nm test light and a 630 nm reference light (pulse rate = 31.25 Hz). The equations were completed while the eye was alternately adapted to a steady 540 or 630 nm light. The adaptation lights were initially adjusted in intensity so that each raised the threshold for a 560 nm test light by an equal amount (0.5 log unit). The reference light and the adaptation lights were produced by using interference filters having half-energy pass bands of 10 nm.

(3) For a subset of the subjects we sought to provide positive evidence for the presence of a short-wavelength sensitive (S) cone. Spectral sensitivity measurements were made under test conditions previously established as effective for achieving this end (Jacobs, Deegan II, & Moran, 1996). These included a stimulus pulse rate of 12.5 Hz, a 460 nm reference light, and continuous adaptation to an intense long-wavelength light that was produced by placing a long-pass filter (50% transmission at 585 nm) in the adaptation beam. Spectral sensitivity measurements were derived from flicker photometric equations made for test lights covering the range from 420 to 510 nm. Other details of the recording were as described above.

3. Results

3.1. Variation in sifaka M/L opsin genes

A total of 10 sifakas (five male, five female) served as subjects for both genetic analysis and the electrophysiological studies. Table 1 shows the variation in the residues detected at position 285 for these animals. Four had Thr at position 285, three had Ala at position 285, and the remaining three animals had both residues. As indicated above, for prosimians like these variation at position 285 predicts corresponding pigment variation. Those predictions are also listed in Table 1. Note that of the female subjects three were heterozygous and are thus predicated to have two different types of M/L cone pigment. An additional nine animals were genetically screened, but not then further tested. Of this group, (four male, five female), seven had only the Thr residue at position 285, while the remaining two were heterozygous at this site (Ala/Thr).

3.2. Test for response univariance

The fast flicker rate and bright lights used for the recording insured that the ERGs reflected cone activity. Under these test conditions we recorded signals from sifakas that were of large amplitude, fully comparable to

Table 1
Comparison of results from genetic screening and spectral sensitivity measurements

Subject	Gender	Residue at position 285	λ_{max} (nm) (Genotype prediction)	λ_{max} (nm) (ERG measurement)
6108	Female	Ala/Thr	543/558	558.3
6333	Male	Thr	558	559.6
6450	Male	Thr	558	559.2
6518	Male	Ala	543	545.3
6583	Male	Ala	543	544.3
6610	Male	Ala	543	546.4
6692	Female	Ala/Thr	543/558	554.2[a]
6723	Female	Thr/Thr	558/558	557.9
6727	Female	Thr/Thr	558/558	557.4
6744	Female	Ala/Thr	543/558	552.3[a]

[a] Diagnosed as trichromatic from the chromatic adaptation test.

those obtained in analogous recordings earlier made on various Old World monkeys and Apes (e.g., Jacobs & Deegan II, 1999; Jacobs et al., 1996).

The test to establish whether the subject had only a single type of M/L cone pigment was conducted subsequent to the other measurements, but since those data serve to organize the spectral sensitivity results they are considered first. For a subject that has only a single type of M/L cone pigment, the ERG photometric equation established for 540 and 630 nm lights is independent of the chromatic adaptation state of the retina. Alternatively, if there is more than one M/L pigment then the equation changes systematically depending on the nature of the chromatic adaptation. The inset in Fig. 1 shows the difference (in log units) in the 540/630 nm photometric equations under the two test conditions for the ten subjects. The results for eight of these animals (triangles) cluster around 0.00, i.e., those equations are independent of adaptation state. The results from the other two animals (circles) show a systematic shift in their photometric equations, requiring more 540 nm light for equation when the eye was adapted to 540 nm light and less 540 nm light when the adaptation was changed to 630 nm. This test has been previously run on many other primate subjects, both human and nonhuman. Human dichromats similarly tested show no significant change in 540/630 nm photometric equations (Jacobs & Deegan II, 1997). On that comparative basis, eight of the sifakas would be predicted to have only a single type of M/L pigment. The remaining two do show systematic change and accordingly would be assumed to have more than one type of M/L cone pigment.

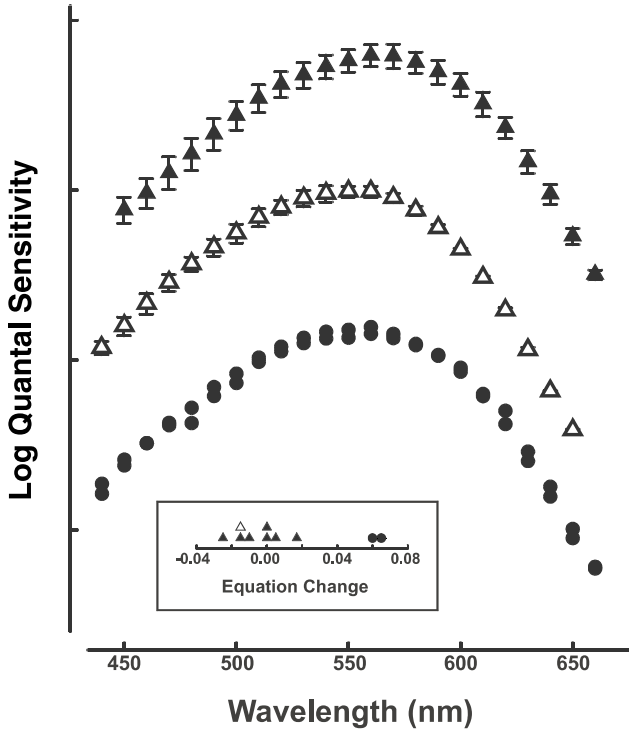


Fig. 1. ERG measurements obtained from three groups of sifakas. The inset at the bottom summarizes the results from a test to establish whether an animal had one or more M/L cone photopigments. Plotted are the differences (given in log units) in the photometric equations obtained between 540 and 630 nm lights under two conditions of chromatic adaptation. Two of the animals (solid circles) had significantly different equations under the two test conditions and thus are inferred to have more than one cone photopigment in the middle to long-wavelengths. The equation values for the other eight animals (triangles) did not change significantly so they have only a single M/L cone pigment. The equation value for an animal whose results differed for the ERG and genetic analyses is indicated by the open triangle (see text). The spectral sensitivity functions for the 10 animals are summarized above. Sifakas having only a single M/L photopigment fall into two groups (closed and open triangles respectively). For those the plotted points are mean sensitivity values (error bars = ±1 SD). Individual results for the animals having two M/L pigments are shown at the bottom. The three sets of results are arbitrarily positioned on the sensitivity axis where each scale division equals 1 log unit.

3.3. Spectral sensitivity

The averaged spectral sensitivity functions for the eight animals predicted to have only a single M/L cone pigment fell into two discrete groups. These are shown in Fig. 1. The majority (five animals plotted at the top as closed triangles) has spectral sensitivity functions that peak at longer wavelengths than do the remainder (three

animals, middle curve, open triangles). As indicated by the error bars, the individual variations in sensitivity for animals in both groups are quite small (for the two groups the standard deviations have averaged values of 0.0714 and 0.0341 log units respectively). The data for each of the two animals that did show a systematic adaptation effect are shown at the bottom of Fig. 1 (solid circles). The sensitivity curves for these animals are consistently broader than for the subjects of the other two groups.

The spectral sensitivity curves for the two subgroups of dichromats were averaged and these data were best fit to photopigment absorption curves (Govardovskii, Fyhrquist, Reuter, Kuzmin, & Donner, 2000) by shifting the curve along the wavelength axis in steps of 0.1 nm to establish the position that yielded the smallest least-squares difference between the data array and the standard curve. Photopigment curves having peak values of 546.5 and 559.7 nm respectively provide good accounts of these data (Fig. 2, top). There are no documented measurements of preretinal absorption in sifakas. Consequently, in deriving the fits of Fig. 2 we assumed that lens density in these diurnal prosimians might be similar to that of another diurnal primate of similar size, the macaque monkey (Boettner, 1967). Whether this choice is reasonable or not, estimation of the peak value is not heavily influenced by assumptions about lens absorption. Thus, truncating the spectral sensitivity functions to include only values for 520 nm and longer, a portion of the spectrum that is relatively uninfluenced by primate preretinal filters (Sharpe et al., 1998), changes the estimated peak locations by about 1.2 nm, in each case shifting them toward the shorter wavelengths. The values obtained for each of the individual animals following this same truncation strategy are entered in Table 1. Estimated in that fashion, the two sifaka M/L photopigments have average peak values of 545.33 nm (SD = 0.91) and 558.48 nm (SD = 1.01). The averaged spectral sensitivity function for the two trichromatic animals has been similarly best fit in Fig. 2 (bottom) by linearly summing these two putative M/L pigment curves in with relative proportions of 543.3 (43.5%) and 558.5 (56.5%).

For two of the animals from the group shown at the top of Fig. 1 it proved possible to also record spectral sensitivity functions under conditions favorable for detecting an S-cone contribution. These results are shown in the form of spectral sensitivity functions in Fig. 3. In each case a significant elevation in sensitivity to the short-wavelengths was detected. The short-wavelength elevation is considerably less prominent for the animal whose results are illustrated at the top, presumably reflecting the fact that for this animal it was only possible

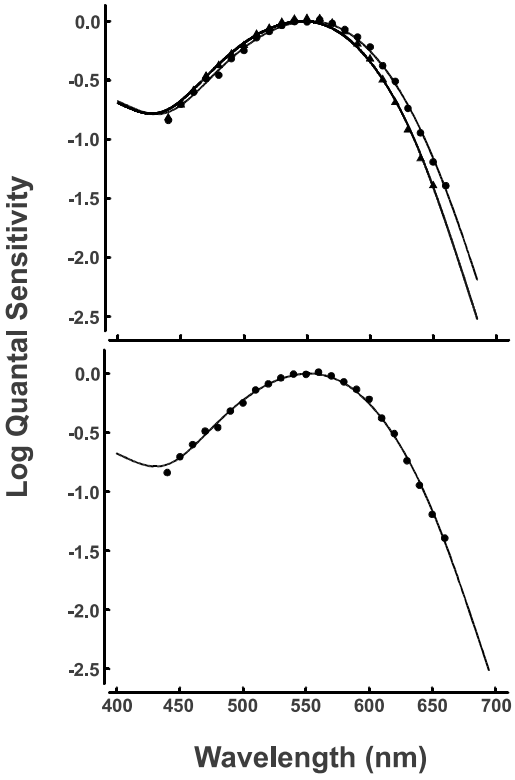


Fig. 2. Cone spectral sensitivity functions for polymorphic forms of sifakas. The curves at the top are averages for the two types of dichromat. The fitted curves have λ_{max} values of 546.5 and 559.7 nm respectively. The average spectral sensitivity function for the two trichromatic animals appears at the bottom. The fitted curve is a linear summation of the two curves from the top. See text for further details.

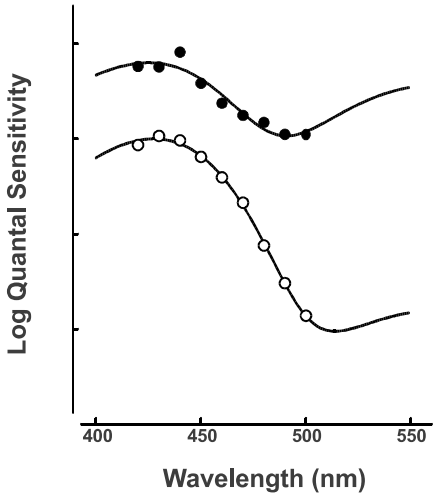


Fig. 3. Spectral sensitivity functions for two sifakas measured using test conditions favorable for detecting contributions from S-cones (see text). The conditions were identical for the two with the exception that long-wavelength adaptation light was approximately 0.3 log unit more intense for the animal whose results are shown at the bottom. The data points have been best fit (continuous line) with a summative combination of two cone pigment templates having peaks and proportions as follows: top [559.6 nm (46.4%) + 427 nm (53.6%)], bottom [559.2 nm (11.3%) + 428 nm (88.7%)]. The two functions have been arbitrarily positioned on the sensitivity axis where each interval equals 0.5 log unit.

to use a concurrent long-wavelength adaptation that was approximately 0.3 log units less intense than could be employed for the second animal (bottom). Both of these animals were potential dichromats whose M/L spectral sensitivity had been measured. The fitted curves in Fig. 3 are accordingly best-fit linear summations of the measured M/L pigment and the putative S pigment. The position of the latter was estimated by searching (in 1 nm steps) to find the spectral position of the S pigment absorption curve that gave the best combined fit. As before, it was assumed that the lens measurements for the macaque monkey provide an acceptable estimate of lens pigmentation in the sifaka. The S pigments positions so determined had λ_{max} values of 427 nm (top) and 428 nm (bottom).

4. Discussion

Coquerel's sifaka, a prosimian primate, has an M/L cone opsin gene polymorphism. This polymorphism yields animals whose retinas contain either or both of two different M/L pigments. The situation is thus very similar to the opsin gene/photopigment polymorphisms found in many platyrrhine monkeys. It has been known for some time that for these primates there is a direct translation from photopigment complement to color vision (Mollon, Bowmaker, & Jacobs, 1984). On those grounds it would seem reasonable to predict that sifakas have either of two types of dichromatic color vision or are trichromatic. The potential incidence of trichromacy among the females will depend on the relative frequencies of the two M/L opsin genes. So far too few sifakas have been examined to evaluate opsin gene incidence although the fact that four of the ten females genotyped were heterozygous would suggest neither of the M/L opsin genes is rare. Of course, color vision requires both combinations of photopigments and a nervous system organization sufficient to appropriately compare photopigment signals. Relatively little is known about prosimian visual systems, but there are grounds for supposing that the neural organizations supporting trichromatic color vision are available. Anatomical examinations of prosimian visual systems have revealed organizational features in the retino-thalamic-cortical pathways similar to those typical of anthropoid primates and, specifically, the presumed substrate for anthropoid M/L cone opponency, P/midget ganglion cells, have been identified (Casagrande & Kaas, 1994; Yamada, Marshak, Silveira, & Casagrande, 1998).

While it seems likely that heterozygous female sifakas have the requisite biology to achieve trichromatic color vision in a formal sense, it is less certain how acute this capacity may be. Although the retina of this species has an *area centralis*, it apparently lacks a fovea (Rohen & Castenholtz, 1967). In line with the lack of a central specialization of the sort typical of all the diurnal anthropoids, the overall cone population in the retina of *Propithecus* is modest with cone density ranging from 34,000 to 12,000 mm/sq^2 along a line connecting the central part of the retina to the periphery (Peichl, Rakotondraparany, & Kappeler, 2001). The peak value is thus only about a sixth of that achieved in the human fovea and a tenth of that observed in the fovea of the New World monkey *Callithrix jacchus* (Goodchild, Ghosh, & Martin, 1996). No other information is available about the nature of the pathways through the retina of the sifaka, but the relative paucity of cones suggests that the capacity for transmitting color information may be lower than that available to trichromatic anthropoids.

With one exception (sifaka 6108, considered below), the results from genetic screening and ERG measurements line up very well. Thus, animals having only a single type of opsin gene tested as dichromatic while those heterozygous for the two M/L opsin genes were judged trichromatic. By extension this suggests that the opsin gene complements of the other prosimian species earlier studied contain similarly strong implications about photopigments (Tan & Li, 1999). As noted above, the residue composition at several critical sites in the opsin can be used to derive predictions about the spectral absorption properties of the M/L pigments. The λ_{max} predictions for the M/L pigments, 543 and 558 nm, are close to the measured spectral sensitivities, but comparison of the rightmost columns of Table 1 show the averaged ERG spectra are slightly longer than the values predicted for each of the pigment positions. That could reflect some incompleteness in the spectral tuning model or a systematic difference between the photopigment absorption properties and the in situ measurement of spectral sensitivity. One factor that can clearly influence the ERG measurements is the presence of intraretinal reflection. *Propithecus*, like many other prosimians, has a tapetum, in this case giving it a prominent golden eye shine (Martin, 1990). Tapetal reflection will provide a relative enhancement of the efficiency of longer wavelength lights (Jacobs & Deegan II, 1992) and potentially this can serve to shift the spectral sensitivity function to a position slightly different than that predicted from the photopigment absorption spectrum. Close inspection of the spectral sensitivity functions of dichromats (Fig. 2) reveals a small elevation just to the right of the peak of the absorption curve. That probably represents a small contribution from tapetal reflection, the net effect of which is to shift the spectral sensitivity functions to slightly longer wavelengths relative to the parent photopigment. As might be expected, the peaks of spectral sensitivity functions for the two diagnosed trichromats (6692, 6744) fall midway between the peaks of the two M/L pigments. The proportions of the two pigment curves required to fit the trichromatic spectral

sensitivity functions of these two were about equal, a fact which implies that their retinas contain about equal numbers of M- and L-cones (Brainard et al., 2000). That result is in accord with the idea that the populations of the two cone types in these heterozygous females are established by random X-chromosome inactivation.

Antibody staining of the retina of *Propithecus* revealed that about 5% of the total cone population was S-cones (Peichl et al., 2001). Their contributions to spectral sensitivity appear in Fig. 3. The estimated spectral peak of the sifaka S-cone, about 427–428 nm, place it in the vicinity of other measured primate S-cones (Calkins, 2001). The indirectness of the measurement technique, however, precludes any strong claims about the exact placement of the sifaka S-photopigment.

The female 6108 stands as an exception to the otherwise close correspondence between the results from genetic screening and ERG measurements. The genetic results indicate she is heterozygous and should have both M/L pigments while the ERG measurements imply she is a dichromat having only the 558 nm photopigment. The genetic result is very compelling since, fortuitiously, 6108 was the mother of two other animals that were tested. These animals (6518 and 6583) were both male and Table 1 indicates that both had the opsin gene for the 543 nm allele and, consequently, so too must 6108. But the ERG results seem equally unambiguous. In addition to a lack of a chromatic adaptation effect (animal indicated by the open triangle in the inset for Fig. 1), the peak of her spectral sensitivity function was indiscriminable from that of the other 558 nm dichromats and significantly longer than that of the trichromats. The implications from the ERG results are made even clearer in Fig. 4 where we have plotted the wavelength-by-wavelength difference in sensitivity between 6108 and the four other 558 nm dichromats. The deviations are extremely small (mean = 0.004 nm) and unsystematic across wavelengths. Shown too in Fig. 4 are similar comparisons of the spectral sensitivity of 6108 and the two identified trichromats. As can be seen, there is a systematic and statistically reliable difference ($p < 0.001$) between her spectral sensitivity and those from comparably tested trichromats. There are, in sum, equally compelling and contradictory indications about the opsin gene and photopigment complements of subject 6108. There is no clear resolution of these conflicting results, but two observations may be relevant. First, this animal was significantly older (18+ years) than any of the other subjects in these experiments and this could conceivably be a factor associated with the absence of expression of any measurable M-cone opsin. Second, there are well documented cases of human subjects who have L/M cone ratios that are so heavily biased that their flicker spectral sensitivity functions appear dichromatic, but who nevertheless have trichromatic color vision (Miyahara, Pokorny, Smith, Baron, & Baron, 1998).

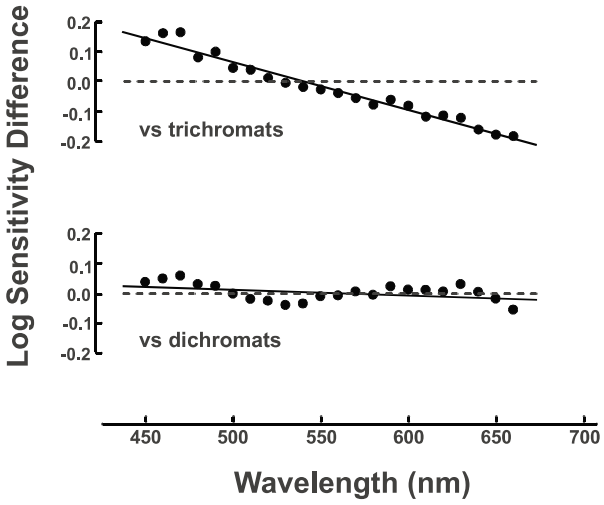


Fig. 4. Comparison of the spectral sensitivity measurements of animal 6108 and other animals. Plotted at the bottom are wavelength-by-wavelength differences in sensitivity between four dichromatic animals and 6108. In computing these differences the curves for 6108 and the dichromatic animals were first adjusted to have the same average sensitivity. The solid line is a linear regression fit to these differences ($r^2 = 0.018$; ns). Above is a similar comparison for the two identified trichromat and subject 6108 ($r^2 = 0.961$; $p < 0.001$).

Results from this single animal aside, this investigation provides clear evidence that the M/L opsin gene polymorphisms recently discovered in prosimians produce a corresponding cone pigment variations. The strong implication to be derived is that trichromatic color vision in primates is not restricted to anthropoids.

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References

Arnason, U., Gullberg, A., & Janke, A. (1998). Molecular timing of primate divergences as estimated by two nonprimate calibration points. *Journal of Molecular Evolution*, *47*, 718–727.

Boettner, E. A. (1967). Spectral transmission of the eye. University of Michigan Contract AF41 (609)-2966. USAF School of Aerospace Medicine, Brooks Air Force Base, Texas.

Boissinot, S., Zhou, Y.-H., Qui, L., Dulai, K. S., Neiswander, H., Schneider, I., Sampaio, I., Hunt, D. M., Hewett-Emmet, D., & Li, W. H. (1997). Origin and molecular evolution of the x-linked duplicate color vision genes in howler monkeys. *Zoological Studies*, *36*, 360–369.

Brainard, D. H., Roorda, A., Yamauchi, Y., Calderone, J. B., Metha, A., Neitz, M., Neitz, J., Williams, D. R., & Jacobs, G. H. (2000). Functional consequences of the relative numbers of L and M cones. *Journal of the Optical Society of America A*, *17*, 607–614.

Calkins, D. J. (2001). Seeing with S cones. *Progress in Retinal and Eye Research*, *20*, 255–287.

Casagrande, V. A., Kaas, J. H. (1994). The afferent, intrinsic, and efferent connections of primary visual cortex in primates. In: A. Peters, K. S. Rockland (Eds.), *Primary visual cortex in primates* (vol. 10, pp. 201–259). New York: Plenum Press.

Goodchild, A. K., Ghosh, K. K., & Martin, P. R. (1996). Comparison of photoreceptor spatial density and ganglion cell morphology in the retina of human, macaque monkey, cat, and the marmoset *Callithrix jacchus*. *Journal of Comparative Neurology*, *366*, 55–75.

Govardovskii, V. I., Fyhrquist, N., Reuter, T., Kuzmin, D. G., & Donner, K. (2000). In search of the visual pigment template. *Visual Neuroscience*, *17*, 509–528.

Jacobs, G. H. (1998). A perspective on color vision in platyrrhine monkeys. *Vision Research*, *38*, 3307–3313.

Jacobs, G. H., & Deegan II, J. F. (1992). Cone photopigments in nocturnal and diurnal procynids. *Journal of Comparative Physiology A*, *171*, 351–358.

Jacobs, G. H., & Deegan II, J. F. (1993). Photopigments underlying color vision in ringtail lemurs (*Lemur catta*) and brown lemurs (*Eulemur fulvus*). *American Journal of Primatology*, *30*, 243–256.

Jacobs, G. H., & Deegan II, J. F. (1997). Spectral sensitivity of macaque monkeys measured with ERG flicker photometry. *Visual Neuroscience*, *14*, 921–928.

Jacobs, G. H., & Deegan II, J. F. (1999). Uniformity of colour vision in Old World monkeys. *Proceedings of the Royal Society of London B*, *266*, 2023–2028.

Jacobs, G. H., Deegan II, J. F., & Moran, J. L. (1996). ERG measurements of spectral sensitivity of common chimpanzees (*Pan troglodytes*). *Vision Research*, *36*, 2587–2594.

Jacobs, G. H., Deegan II, J. F., Neitz, J. A., Crognale, M. A., & Neitz, M. (1993). Photopigments and color vision in the nocturnal monkey, Aotus. *Vision Research*, *33*, 1773–1783.

Jacobs, G. H., & Neitz, J. (1987). Inheritance of color vision in a New World monkey (*Saimiri sciureus*). *Proceedings of the National Academy of Sciences USA*, *84*, 2545–2549.

Jacobs, G. H., Neitz, J., & Krogh, K. (1996). Electroretinogram flicker photometry and its applications. *Journal of the Optical Society of America A*, *13*, 641–648.

Jacobs, G. H., Neitz, M., Deegan II, J. F., & Neitz, J. (1996). Trichromatic colour vision in New World monkeys. *Nature*, *382*, 156–158.

Jacobs, G. H., Neitz, M., & Neitz, J. (1996). Mutations in S-cone pigment genes and the absence of colour vision in two species of nocturnal primate. *Proceedings of the Royal Society of London B*, *263*, 705–710.

Kainz, P. M., Neitz, J., & Neitz, M. (1998). Recent evolution of uniform trichromacy in a New World monkey. *Vision Research*, *38*, 3315–3320.

Martin, R. D. (1990). *Primate origins and evolution*. Princeton: Princeton University Press.

Miyahara, E., Pokorny, J., Smith, V. C., Baron, R., & Baron, E. (1998). Color vision in two observers with highly biased LWS/MWS cone ratios. *Vision Research*, *38*, 601–612.

Mollon, J. D., Bowmaker, J. K., & Jacobs, G. H. (1984). Variations of colour vision in a New World primate can be explained by polymorphism of retinal photopigments. *Proceedings of the Royal Society of London B*, *222*, 373–399.

Nathans, J., Thomas, D., & Hogness, D. S. (1986). Molecular genetics of human color vision: the genes encoding blue, green and red pigments. *Science*, *232*, 193–202.

Neitz, M., Neitz, J., & Jacobs, G. H. (1991). Spectral tuning of pigments underlying red–green color vision. *Science*, *252*, 971–974.

Peichl, L., Rakotondraparany, F., & Kappeler, P. (2001). Photoreceptor types and distributions in nocturnal and diurnal Malagasy primates. *Investigative Ophthalmology and Visual Science*, *42*, S48.

Rohen, J. W., Castenholtz, A. (1967). Über die Zentralisation der Retina bei Primaten. Folia Primatologica 5, 92–147.

Sharpe, L. T., Stockman, A., Jagle, H., Knau, H., Klausen, G., Reitner, A., & Nathans, J. (1998). Red green and red–green hybrid pigments in the human retina: correlations between deduced protein sequences and psychophysically measured spectral sensitivities. *Journal of Neuroscience*, *18*, 10053–10069.

Shyue, S.-K., Boissinot, S., Schneider, H., Sampaio, I., Schneider, M. P., Abee, C. R., Williams, L., Hewett-Emmett, D., Sperling, H. G., Cowing, J. A., Dulai, K. S., Hunt, S. M., & Li, W.-H. (1998). Molecular genetics of spectral tuning in New World monkey color vision. *Journal of Molecular Evolution*, *46*, 697–702.

Tan, Y., & Li, W.-H. (1999). Trichromatic vision in prosimians. *Nature*, *402*, 36.

Yamada, E. S., Marshak, D. W., Silveira, L. C. L., & Casagrande, V. A. (1998). Morphology of P and M retinal ganglion cells of the bush baby. *Vision Research*, *38*, 3345–3352.